

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Examination Summer – 2024

Course: B. Tech. Branch: Mechanical Engineering / Mechanical Engineering (Sandwich) / Mechanical and Automation

Subject Code & Name: BTMC303 Thermodynamics

Semester :III

Max Marks: 60

Date: 04/07/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. Attempt any two sub questions from each question
3. Use of non-programmable scientific calculators is allowed.
4. Illustrate your answer with neat sketches, diagram, etc., wherever necessary.
5. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Explain different kinds of thermodynamic systems with real-life examples.	Remembering	6
B) The density of a liquid varies with elevation as $\rho = \rho_o(1 + z)^{0.25}$ where $\rho_o = 950 \text{ kg/m}^3$ at the free surface of the liquid. Determine the pressure difference between a point at 10 m deep in the liquid column and the free surface.	Analyzing	6
C) i) State zeroth law of thermodynamics.	Understanding	3+3
ii) Explain precisely the relationship between state and property of the system.		
Q.2 Solve Any Two of the following.		12
A) State first Law of thermodynamics for a cyclic process and change of state for the closed system defining each term.	Remembering	6
B) Apply SFEE in a Nozzle and obtain the expression of exit velocity in terms of enthalpies difference applying suitable assumptions.	Understanding	6
C) A water heater operates under steady flow conditions receiving 227kg/min of water at 74 °C and enthalpy 308 kJ/kg. The water is heated by mixing with steam which is supplied to the heater at 102 °C, enthalpy 2680 kJ/kg. The mixture leaves the heater as liquid water at 100 °C, enthalpy 420 kJ/kg. How many kg of steam per hour must be supplied to the heater?	Analyzing	6
Q. 3 Solve Any Two of the following.		12
A) Show that the shape of the Carnot cycle is rectangular on T-s diagram and obtain its expression in terms of source and sink temperatures.	Understanding	6
B) Show that entropy of the Universe always increases.	Understanding	6
C) The condenser of a room air-conditioning unit rejects heat to the ambient at the rate of 2.5 kW. The compressor of the same unit consumed 1 kW power. Determine the refrigerating effect produced by the air-conditioning unit and its COP.	Analyzing	6

Q.4 Solve Any Two of the following.		12
A) Define ideal gas. What is Universal gas constant? Obtain its value.	Remembering	6
B) A certain gas has $c_p = 0.913$ and $c_v = 0.653$ kJ/kg K. Find the molecular weight and the gas constant R of the gas.	Analyzing	6
C) Show that for an ideal gas $c_p - c_v = R$.	Analyzing	6
Q. 5 Solve Any Two of the following.		12
A) Explain the working of the throttling calorimeter with the schematic diagram and process representation on h-s diagram.	Remembering	6
B) Find the enthalpy and entropy of steam when the pressure is 1.4 MPa, 380 °C	Analyzing	6
C) Define the following; Saturated steam, wet steam, dryness fraction	Remembering	6

END